

Writing Tips and More

Guidance for Home Assignments in Econometrics A

Bertel Schjerning

Department of Economics

University of Copenhagen

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Abstract

This note offers practical advice on writing econometrics assignments: structuring a report, presenting linear regression models and assumptions, explaining empirical results and simulations, and producing clear tables, figures, and reproducible code. Apply the guidance to the methods and questions relevant to each assignment, following its specific instructions.

1 Writing and structure

1.1 Start with the assignment

Read the entire assignment before you begin and answer all questions. Follow its instructions on report length, formatting, group work, assessment, and submission. The advice below concerns how to present your work; include theory, analysis, and discussion relevant to the questions asked.

1.2 Make the report readable on its own

Presume general knowledge of econometrics, but no prior knowledge of the particular assignment. Write for a student who has completed the course. It should be clear **which question you address, how you address it, and what you conclude**.

- Explain what you have done and why, so that the reader can understand and reproduce your empirical analysis.
- Write in precise, easy-to-understand academic language.
- Be compact. Include what is needed to make the argument, and avoid repetition.
- Use correct and consistent notation. Proofread for typos.
- Present theory and results in a logical order.

1.3 Choose a clear structure

Use headings and, where helpful, question numbers to make every answer easy to find. Following the order of the assignment is often a good starting point. A report will typically introduce the question and data, present the relevant econometric theory and empirical results, and end with a conclusion.

These elements need not be separate sections. In a short report, it may be clearer to discuss each model and its results together. Let the questions and page limit determine how much space each element needs.

1.4 Keep the introduction short

Briefly explain:

1. **Question:** What is the main question or set of questions?
2. **Motivation:** Why is it of interest?
3. **Approach:** Which data and econometric models or methods do you use?
4. **Main finding:** What do you conclude?

Introduce the analysis rather than reproducing the assignment text. A few focused sentences may be enough.

2 Linear regression: theory and assumptions

2.1 State and interpret the model

Write the regression equation before discussing it. Define the variables, parameters, observational unit, and any time or group indices. Explain units, log transformations, squared terms, and the reference categories for dummies. A model can be linear in its parameters even when regressors are transformed.

Interpret coefficients in the stated specification: explain what is held constant, distinguish percentages from percentage points, and account for squared terms or interactions when relevant. For an effect involving several coefficients, identify the combination rather than interpreting one coefficient in isolation.

2.2 Estimator, derivation, and properties

- State the estimation method used: OLS, WLS, fixed effects, or IV/2SLS, as relevant to the question. Some of these methods may not have been introduced yet; discuss only those relevant to the current assignment. Explain any weights, transformations, or instruments where applicable.
- When asked to derive a result, state the starting equation and show the intermediate steps. In matrix arguments, define the matrices and check their dimensions. Pay attention to intercepts, centering, and full-rank conditions.

- Explain which assumptions give unbiasedness, consistency, or efficiency. Keep finite-sample properties distinct from asymptotic properties.
- Follow the assumptions stated in the question. Distinguish assumptions you are asked to maintain from those you are asked to discuss or introduce.
- Use consistent notation throughout. Adapt formulas from slides or notebooks to your own variables and parameters.

An empirical comparison can illustrate a theoretical result, but does not replace a requested proof. Include relevant theory, not a general textbook chapter.

2.3 Explain the mechanism behind an assumption

Avoid “there could be omitted variables” without saying which variables and why they matter. For example, in a wage regression with education:

“Unobserved worker ability may affect productivity and hence wages, so it belongs in the error term. If ability also affects educational attainment, education may be correlated with the error term.”

This links the economic argument to the regressor and error term. Give enough detail to explain the mechanism, without a lengthy hypothetical story.

If instruments are relevant to your assignment (a topic introduced later in the course), discuss **relevance and exogeneity separately**. Explain the economic argument for each; a significant first stage does not by itself establish exogeneity. Keep causal interpretations consistent with the identifying assumptions.

3 Empirical analysis, inference, and simulation

3.1 Describe the data and the specifications

Identify the sample, observational unit, period, and variable measurements. Use relevant descriptive statistics and figures, and comment on their main features. Explain sample restrictions and constructed variables.

Label each estimated specification clearly. State which controls, dummies, weights, or transformations are included. Distinguish models prescribed by the question from choices you make yourself. Report the requested intermediate models as well as the final specification.

3.2 Interpret estimates and comparisons

- Discuss magnitudes, signs, and uncertainty, and relate results to your predictions or economic theory. A table alone does not answer an interpretation question.
- When comparing estimates, identify what changes in the sample, specification, or estimation method, and explain the mechanism behind any differences.

- Label goodness-of-fit measures and define sums-of-squares notation consistently. Consider whether a transformation changes the dependent variable or the variation being explained before comparing R^2 across regressions.

3.3 Make hypothesis tests explicit

State H_0 and H_1 in terms of model parameters, identify the test and its assumptions, and report the statistic, reference distribution (with relevant degrees of freedom), and p -value or critical value. Give the statistical conclusion and its meaning for the question. Failure to reject is not proof that the null is true.

State how standard errors are computed, including whether they are heteroskedasticity-robust. Distinguish changing the standard-error calculation from changing the estimator. For linear combinations of coefficients, use the relevant covariance terms. When testing model assumptions or instrument validity, discuss what the test can and cannot establish.

3.4 Explain Monte Carlo experiments

State the data-generating process, parameter values, sample size, number of replications, and scenarios. Explain what is generated and estimated in each replication. Use a reproducible random seed and implement the specified conditional error variance carefully.

Label summaries clearly: distinguish bias, the standard deviation of estimates across replications, the average estimated standard error, and rejection frequencies. State the null and significance level for simulated tests. Compare the results with theoretical predictions and explain how changes in sample size or scenario affect them.

3.5 Conclude on the question

Give a brief answer supported by the reported evidence. Discuss relevant limitations and assumptions. Additional tests or literature comparisons should serve the task; they are not automatic requirements for every assignment.

4 Presenting results and submitting code

4.1 Tables and figures

- **Show relevant results.** Include the requested coefficient estimates and standard errors, along with other quantities needed to understand the analysis, such as sample sizes or relevant test results.
- **Make labels informative.** Number tables and figures, provide captions, and use clear variable and model names. State the dependent variable, units, transformations, and differences between specifications where relevant.
- **Explain the notation.** Table notes should state what is in parentheses and define any significance stars or other symbols. Check that stars agree with the reported results.

- **Use sensible precision.** Avoid the eighth decimal place, but retain enough digits to make the results and comparisons meaningful.
- **Format the output.** Use readable tables and figures rather than screenshots or raw Python output. Leave out irrelevant software diagnostics.
- **Discuss the evidence.** Refer to each table or figure in the text and explain what the reader should learn from it.

Do not omit relevant results, such as estimates for parameters needed to understand a requested model. Equally, do not include every statistic printed by the software. Choose content that helps the reader follow the analysis.

4.2 A reproducible Jupyter Notebook

Prepare one organized notebook that generates all tables and figures in the report, in the same order. Comments should clearly identify the corresponding table or figure. Put all group members' names and study IDs in both the notebook and the report's front page.

- Organize the notebook with headings and comments that follow the report.
- Use the supplied data and a clearly documented path. Avoid paths that only work on your own computer.
- Include the necessary code in the notebook. Do not rely on extra student-written files unless the submission instructions allow them.
- Before submission, restart the kernel and run every cell in order. The notebook should execute without errors and reproduce the results in the PDF.
- Use software consistently with the assignment instructions. Understand and explain the methods you apply; software output does not replace requested algebra or interpretation.

4.3 Final check

Have you answered every question? Are the notation, tables, and written conclusions consistent? Does the report follow the required page limit and format? Are all names and study IDs included? Submit the PDF and notebook as directed, once per group, and check that the uploaded files are the final versions.